DEPARTMENT OF COMPUTER SCIENCE

MASTER DEGREE IN COMPUTER SCIENCE

PROJECT REPORT
IN
Semantics in Intelligent Information Access

FAIR ISLE — Detect and Rephrase non-inclusive texts

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Abstract

FAIR ISLE is a webapp that identifies non-inclusive language and proposes inclusive rewrites in Italian and English. The project combines a bias classifier with a seq2seq rewriter trained on curated datasets expanded through data augmentation, under psychological supervision. The pipeline integrates a user feedback cycle for continuous improvement and defines a multi-level evolutionary roadmap to increase contextual sensitivity. Quantitative results indicate high performance in monolingual classification and a rewriting quality consistent with the adopted automatic metrics.

Executive Summary

FAIR ISLE is an inclusive-language assistant delivered as a web application. Given an input text, the system (i) detects non-inclusive sentences and (ii) proposes inclusive rewrites that preserve meaning, tone, and communicative intent. The project targets both Italian and English, and also explores a multilingual setting to evaluate cross-lingual generalization.

The solution is implemented as a modular pipeline composed of a bias classifier and a sequence-to-sequence rewriter. To ensure interactive response times, the rewriting module is activated only when the classifier predicts non-inclusivity with sufficient confidence. A production threshold of `0.60` is used to separate non-relevant cases from actionable ones. The backend is implemented in Flask and relies on Hugging Face Transformers pipelines with locally stored models.

Data quality and safety were treated as first-class concerns. All dataset operations—from initial cleaning to large-scale data augmentation—were performed under formal psychological supervision, ensuring that rewrites remain not only linguistically correct but also emotionally appropriate and non-paternalistic. The Italian and English datasets were developed as native resources (not simple translations) to reduce cultural transfer of irrelevant biases.

Quantitatively, the classifiers achieved very high performance across settings (approximately 99% accuracy and F1-score). Rewriter evaluation followed a multi-metric protocol (ROUGE, BLEU, BERTScore) to capture lexical overlap, precision, and semantic similarity. Specialist monolingual models (English and Italian) slightly outperformed the multilingual rewriter, confirming the expected trade-off between specialization and generalization.

Finally, FAIR ISLE includes an active feedback loop. Users can validate the detection outcome and rate or correct suggested rewrites; feedback is stored in a

JSON repository and enables future retraining cycles. This design supports continuous improvement and positions the tool as an evolving assistant aligned with real usage patterns.

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Capitolo 1

Introduction

1.1 The context: Language and social perception

Language is not merely a tool to describe reality, but a means that actively contributes to shaping it, influencing attitudes, behaviors, and social norms. As highlighted by the European Institute for Gender Equality (EIGE), words matter in defining our worldview: the predominant use of masculine terms for general references reflects and reinforces assumptions about gender roles, contributing to the invisibility of women in the public sphere. Inclusive or gender-sensitive language, therefore, is not a stylistic whim, but a strategy to make differences and needs visible, reduce unconscious stereotypes, and promote equal opportunities.

1.2 Regulations and Reference Guidelines

The development of tools for linguistic inclusivity fits within a well-defined regulatory and cultural framework, both internationally and nationally. At the international level, UN Women and the European Union promote guidelines to avoid discriminatory formulations and encourage neutral or doubly marked terms when relevant. The American Psychological Association (APA) insists on specificity and neutrality in describing people, avoiding demeaning labels and unnecessary binaries. In Italy, primary references are the MIUR Guidelines (2018) and UNI standards for gender equality in administrative and educational language.

1.3 Current linguistic challenges

From the state of the art, three classes of problems emerge that a rewriting system must address:

- **Stereotypes:** rigid attribution of roles or abilities based on gender (e.g., care professions associated with women).
- **Invisibility and omission:** use of the overextended masculine or generic terms that erase female presence.
- **Subordination:** use of suffixes or asymmetrical forms of address that imply lesser value or infantilization.

1.4 Goal of the FAIR ISLE tool

FAIR ISLE was created with the objective of combining linguistic accuracy, computational efficiency, and social usefulness:

- Automatically detect non-inclusive expressions in short and long texts.
- Generate inclusive rewrites that preserve meaning, tone, and communicative intent.
- Support Italian and English, with experimentation of a multilingual version.
- Integrate a feedback loop to collect human corrections.
- Ensure response times suitable for an interactive webapp.

1.5 Implemented Linguistic Methodology

Rewriting is not limited to grammatical corrections, but applies strategies that reduce bias while maintaining text readability:

- **Double inflection:** explicit mention of both genders (e.g., “tutte e tutti”).
- **Paraphrasing:** replacement of the overextended masculine with neutral forms (e.g., “personale”, “chi ha interesse”).

- **Impersonal or passive forms:** shifting the focus to the action or the object to remove gender reference.

In tests, double inflection increased visibility but can weigh down the text; for this reason, paraphrases and impersonal forms are preferred when they do not compromise clarity and naturalness.

Capitolo 2

Technical Architecture and Operation

2.1 System overview

The system is organized as a modular pipeline composed of classification and rewriting, with confidence handling and structured output. However, during the development phase, the limits of this "flat" approach were identified: it does not always capture ambiguous expressions and lacks contextual memory. The backend is implemented in Flask and uses Hugging Face Transformers pipelines with models saved locally in `models/`. In production, the confidence threshold is set to `0.60` to distinguish non-relevant cases, while the UI highlights the sentence status and the generated suggestion.

2.2 Analysis pipeline

- Segmentation of the text into sentences.
- Bias classification with associated probability.
- If confidence is low, label as "not relevant"; if the sentence is non-inclusive, trigger the rewriter.
- Return of label, score, and rewrite proposal to the interface.

Capitolo 3

The Dataset and Its Evolution

The beating heart of the FAIR ISLE project is represented by the knowledge base on which the model was trained. The quality and variety of the data are decisive to ensure that the system not only recognizes explicit forms of discrimination, but also captures subtle nuances of non-inclusive language.

A fundamental and cross-cutting aspect of all the work on data was rigorous scientific supervision: every operation carried out on the datasets, from initial cleaning to massive generation of new examples, was supervised and received formal approval from the team psychologist. This ensured that linguistic corrections were not only grammatically correct, but also psychologically and emotionally appropriate, avoiding the risk of replacing one bias with another or using paternalistic language.

3.1 The initial dataset: structure and cleaning

The starting point was the file `dataset_chatgpt_001.xlsx`, an initial corpus composed of 449 examples. This dataset was built by collecting sentences representative of different categories of bias: gender stereotypes, discrimination based on age, ethnicity, sexual orientation, and disabilities.

Each entry in the original dataset was structured into multiple fields: the "non-inclusive sentence", the "inclusive sentence" (the proposed correction), a possible alternative variant, the identified bias type, and a validation field. Before use, a rigorous cleaning operation was carried out: only positively validated rows were

selected, discarding doubtful or non-relevant examples. In this phase, the psychologist's review was decisive in choosing the most sensitive and empathetic rewrite alternatives. The result was a clean data core, free of duplicates and semantically consistent.

3.2 Data augmentation and Italian dataset

To increase variety, data augmentation was performed with Google Gemini, generating new variations consistent with the original corrections. The first extended dataset (`dataset_completo`) reached 5,367 pairs. Subsequently, the Italian corpus was expanded further up to `dataset_it.json` (16,672 pairs), used for fine-tuning the IT5 rewriter.

3.3 English dataset

In parallel with work on Italian, a specific dataset for English was developed, leveraging the generative capabilities of Google Gemini. The goal was not a literal translation of the Italian dataset—which could import non-relevant cultural biases—but the creation of a native corpus for the anglophone context. Google Gemini was instructed on international guidelines (such as those of the UN or the APA) to generate from scratch 4,980 sentence pairs. Examples addressed the same macro-topics (gender bias, ageism, ableism) using English-specific constructions such as *singular they* and culturally grounded stereotypes. The psychologist's final approval certified that the inclusive and empathetic intent was preserved and correctly transferred into English.

3.4 Multilingual dataset

The Italian and English datasets were unified into `dataset_multi.json` (21,653 pairs), normalizing JSON Lines format with the fields `non_inclusiva` and `inclusiva`. Mixing languages forced models to generalize the concept of bias beyond language-specific patterns.

Capitolo 4

Implementation of AI Models for the English Language

The FAIR ISLE architecture for English follows a dual structure: a classifier for bias detection and a seq2seq rewriter for inclusive transformation. Development, training, and validation were conducted in Google Colab to leverage cloud GPU resources and ensure reproducibility through Jupyter notebooks.

4.1 The Bias Classifier

4.1.1 Model choice: DistilBERT

For this task, `distilbert-base-uncased` was selected to balance performance and inference latency. DistilBERT retains over 95% of BERT performance while being smaller and faster, which is crucial for real-time web usage.

4.1.2 Data preparation and training

1. **Labeling:** 0 for inclusive text, 1 for non-inclusive.

2. **Tokenization:** `AutoTokenizer` with truncation for variable lengths.

3. **Training:** Hugging Face `Trainer`, 3 epochs, learning rate `2e-5`, batch size 16 (train) and 32 (eval), weight decay `0.01`.

4. **Evaluation and saving:** evaluation at each epoch; save best model using F1-score; enable mixed precision (FP16).

4.2 The Inclusive Rewriter

4.2.1 Model choice: Google FLAN-T5

For rewriting, `google/flan-t5-base` was selected. T5 is a text-to-text architecture designed for seq2seq tasks; the FLAN variant benefits from instruction tuning, improving compliance with complex prompts such as rewriting inclusively without changing meaning.

4.2.2 Data preparation and training

- **Preprocessing and prompting:** tokenize pairs with `max_length=128`; each input prefixed by:

  ```
  Rewrite the sentence using inclusive language:
  ```

- **Fine-tuning:** 3 epochs, learning rate `5e-5`, warmup ratio 5%, batch size 4 with gradient accumulation steps 2.
- **Evaluation (multi-metric):** ROUGE, BLEU, and BERTScore.
- **Saving:** checkpoint each epoch, keep only the 2 best checkpoints.

4.3 Model Integration

In operational mode, the two models work sequentially:

1. user input → DistilBERT classifier;
2. if output is *Non-Inclusive*, forward to FLAN-T5 rewriter;
3. if output is *Inclusive*, return positive feedback without changes.

This modular approach maximizes efficiency and specialization.

Capitolo 5

Implementation of AI Models for the Italian Language

The Italian pipeline mirrors the English dual structure but required language-specific strategic choices and infrastructure. Classifier training was performed on Google Colab, while the more demanding rewriter training was conducted on Kaggle GPUs.

5.1 The Bias Classifier

5.1.1 Model choice: BERTino

For Italian, `indigo-ai/BERTino` was selected as a monolingual BERT model pre-trained on large Italian corpora, offering stronger linguistic coverage than generic multilingual models.

5.1.2 Data preparation and training

- **Preprocessing:** labels (0 inclusive, 1 non-inclusive); `AutoTokenizer` with `truncation=True`.
- **Training:** 3 epochs, batch size 16 (train) and 32 (eval), learning rate `2e-5`, weight decay `0.01`.
- **Evaluation:** evaluate each epoch; `load_best_model_at_end=True`.

- **Metric:** F1-score as `metric_for_best_model`; mixed precision `fp16`.

5.2 The Inclusive Rewriter

5.2.1 The initial challenge and the change of strategy

An initial attempt used a multilingual FLAN-T5 variant, but preliminary tests revealed unnatural output with heavy redundancies and repetitions, undermining readability. The model struggled with Italian morphological complexity.

5.2.2 The final solution: IT5

The strategic decision was to adopt `gsarti/it5-base` (IT5), a T5 model pre-trained specifically for Italian text-to-text tasks.

5.2.3 The importance of the extended dataset

The new model made it evident that the initial dataset was insufficient for robust generative training. This motivated the large-scale data augmentation that produced the extended Italian corpus (`dataset_it.json`, 16,672 pairs), providing sufficient variety to learn stable inclusive rewriting patterns.

5.2.4 Data preparation and training on Kaggle

- **Prompting:** tokenize with `max_length=128`; each input prefixed by:

  ```
  [INCLUSIVO] Riformula la frase in modo inclusivo e rispettoso,
  mantenendo il tema...
  ```

- **Fine-tuning:** 3 epochs with `Seq2SeqTrainer`, learning rate `5e-5`, warmup ratio 5%, batch size 4 with gradient accumulation steps 2, mixed precision `fp16`, and `predict_with_generate`.

- **Evaluation:** ROUGE, BLEU, and BERTScore.

Cross-analysis confirmed a clear improvement in naturalness and correctness compared to initial FLAN-T5 attempts.

Capitolo 6

Implementation of AI Models for the Multilingual Setting

6.1 Unification and expansion of the dataset

To overcome system fragmentation, the goal became a language-agnostic architecture operating on mixed inputs without preselecting the language. Italian and English resources were merged into `dataset_multi.json` with normalized JSON structure (`non_inclusiva`/`inclusiva`), producing a corpus of over 21,000 examples. This mixed dataset forces learning the abstract concept of bias beyond single-language patterns.

6.2 The Multilingual Classifier

6.2.1 Model choice: XLM-RoBERTa

The multilingual classifier uses `xlm-roberta-base`, a cross-lingual model pre-trained on 100 languages. Its aligned representations help generalize semantic patterns of bias across languages.

6.2.2 Training protocol

1. **Preprocessing:** labels (0 inclusive, 1 non-inclusive); `AutoTokenizer` with `truncation=True` (SentencePiece-based).

2. **Fine-tuning:** shuffled bilingual batches; 3 epochs; batch size 16 (train) and 32 (eval); learning rate `2e-5`; weight decay `0.01`.

3. **Evaluation:** epoch-based evaluation; save best model with weighted F1-score; mixed precision `fp16`.

6.3 The Multilingual Rewriter

6.3.1 Model choice: Google mT5

For multilingual rewriting, `google/mt5-base` was selected. mT5 is trained on mC4 across 101 languages, making it suitable for cross-lingual text-to-text rewriting.

6.3.2 Training on Kaggle

- **Tokenization:** mT5 tokenizer with `use_fast=False` for stability on multilingual special characters; `max_length=128`.
- **Optimizer:** Adafactor to reduce memory footprint.
- **Batching:** batch size 2 with gradient accumulation steps 4; gradient checkpointing enabled.
- **Training:** 3 epochs; learning rate `5e-5`.
- **Evaluation:** ROUGE, BLEU, and BERTScore.

Capitolo 7

Quantitative Performance Analysis

Validating a Natural Language Generation system is complex: unlike classification (right/wrong), rewriting can be correct in multiple ways. To evaluate FAIR ISLE models (English, Italian, Multilingual), a testing pipeline combines ROUGE, BLEU, and BERTScore.

7.1 Computation methodology

For each flow:

1. **Diagnostic analysis:** accuracy in distinguishing inclusive vs non-inclusive sentences.

2. **Generative analysis:** compare generated rewrites vs human-validated ground truth using ROUGE, BLEU, BERTScore.

For each rewriter (FLAN-T5, IT5, mT5), the protocol is:

1. Generation on unseen non-inclusive sentences.

2. Comparison against ground truth references.

3. Metric computation with standard algorithms.

Classifier	**Accuracy**	**F1-Score**
English (DistilBERT)	99.70%	0.997
Italian (BERTino)	99.25%	0.992
Multilingual (XLM-RoBERTa)	99.32%	0.993

Tabella 7.1: Classifier performance across language settings.

7.2 Classifier results

Why such high values? Results above 99% are rare in complex NLP tasks, but here they can be explained by:

1. **Pre-trained knowledge:** fine-tuning focuses attention on discriminatory patterns rather than teaching the language.

2. **Bias specificity:** non-inclusive language often follows recurring, strong lexical/syntactic cues.

3. **Dataset quality:** rigorous cleaning and human validation reduced noise and improved learnability.

7.3 Rewriter results

Rewriter	**ROUGE-1**	**BLEU**	**BERTScore F1**
English (FLAN-T5)	~35.94	~10.42	0.899
Italian (IT5)	~23.27	~5.98	0.886
Multilingual (mT5)	~21.51	~6.00	0.885

Tabella 7.2: Rewriter performance across language settings.

7.3.1 Interpretation

English rewriter: the high BERTScore indicates strong semantic preservation while removing bias.
Italian rewriter: ROUGE/BLEU are lower due to Italian's richer morphology and flexible word order, but BERTScore remains high, confirming semantic fidelity.

Multilingual rewriter: results are close to Italian but slightly lower overall; generation consistency can be weaker than specialist models.

Capitolo 8

Comparative Analysis and Final Considerations

A clear trend emerges: monolingual (specialized) models tend to outperform the multilingual (generalist) model, with different margins depending on the task.

8.1 Higher quality of specialist models

Reasons for the gap:

1. **Curse of multilinguality:** mT5 must distribute capacity across many languages, reducing per-language specialization.
2. **Tokenization compromises:** multilingual vocabularies often split words more, increasing generation difficulty.
3. **Task complexity:** multilingual rewriting must both infer the language and apply inclusive correction.

Capitolo 9

User Interface and User Experience (UX)

9.1 The prototyping phase: pure functionality

Early development focused on validating model capabilities, so the initial interface was extremely essential, with basic input/output fields and limited visual identity.

9.2 The redesign: modernization and visual identity

As the backend matured, the frontend was fully redesigned to align the visual experience with technological quality, introducing a modern, clean, professional graphic language.

9.3 Active feedback loop: the user as validator

The UI introduces a double evaluation channel:

- confirmation or denial of classification (reducing false positives and false negatives);
- evaluation of rewriting with the possibility to propose an alternative correction;

- storage of feedback in a JSON database for future retraining cycles.

Capitolo 10

Project Info

- **Exam:** Semantics in Intelligent Information Access
- **Project:** FAIR ISLE — Inclusive-Language Detection & Rewriting
- **GitHub:** `https://github.com/Roberto1512/SIIA`

Essential references

- European Institute for Gender Equality (EIGE) — guidelines on gendered language.
- UN Women and the European Union — guides for inclusive communication.
- American Psychological Association (APA) — Bias-Free Language Guidelines.
- MIUR (2018) — Guidelines for the use of gender in administrative language.
- UNI — guidelines for gender equality in language.
- Hugging Face Transformers and models DistilBERT, BERTino, FLAN-T5, IT5, XLM-RoBERTa, mT5.